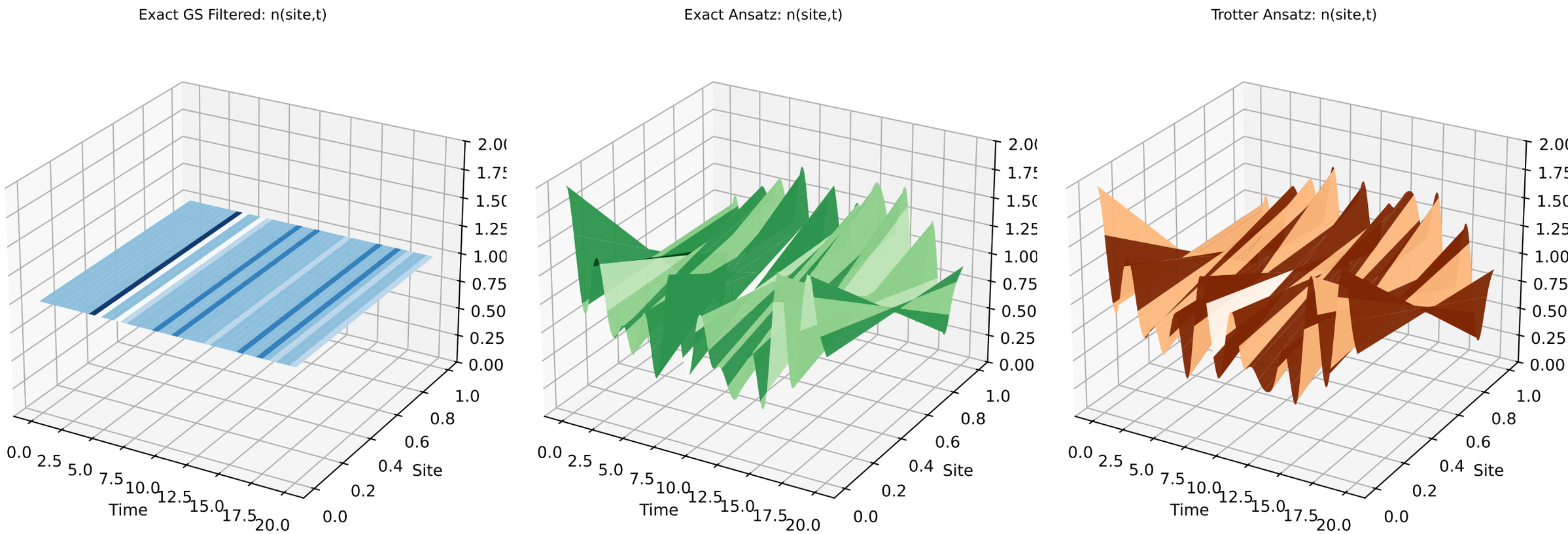
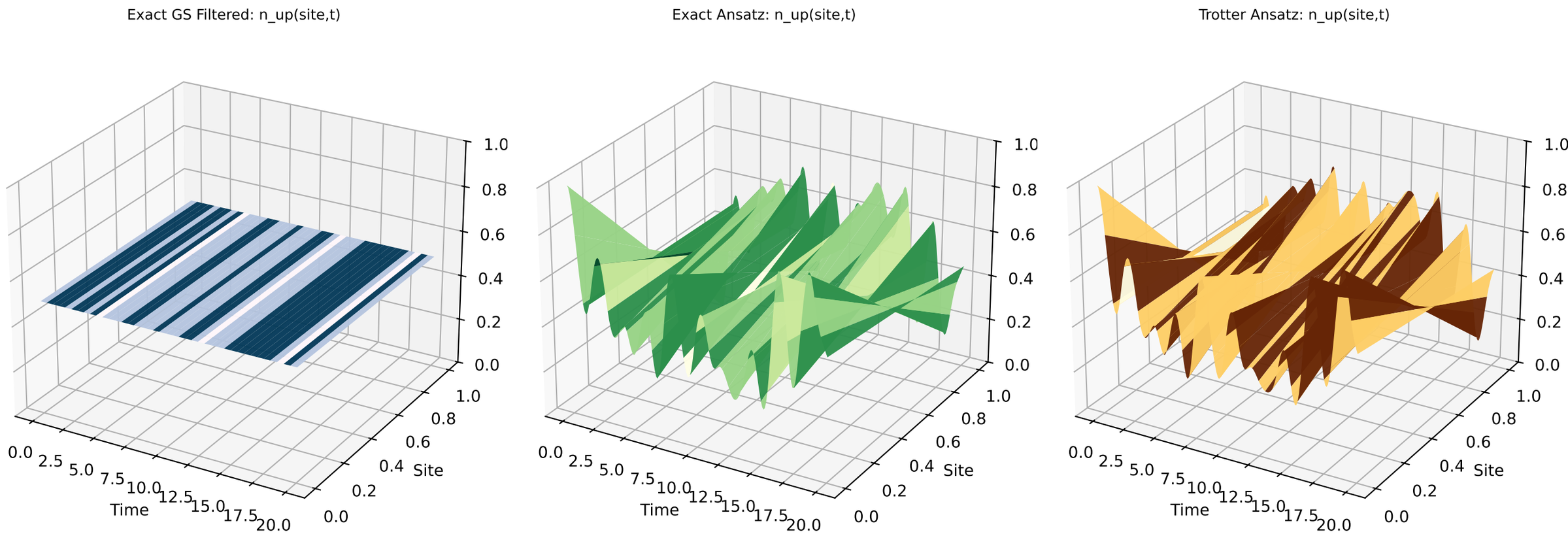
```
Parameter Manifest — Hubbard-Holstein L=2

Model:
  - model_family: Hubbard-Holstein
  - problem: hh

Ansatz:
  - ansatz_type: hh_hva_tw
  - vqe_method: hardcoded_hh_hva_termwise_statevector

Drive Enabled: False

Core Physical Parameters:
  - t=1.0
  - U=2.0
  - dv=0.0

Hubbard-Holstein Parameters:
  - omega0=1.0
  - g_ep=1.0
  - n_ph_max=1
  - boson_encoding=binary
  - nq_fermion=4
  - nq_phonon=2
  - nq_total=6

Trajectory Settings:
  - t_final=20.0
  - num_times=201
  - trotter_steps=64
  - suzuki_order=2
  - term_order=sorted
  - boundary=periodic
  - ordering=blocked
```

```
  - initial_state_source=hf

Topline Results:
  - subspace_fidelity_at_t0: 0.1551829166136877
  - vqe_energy: -0.3895416454240122
  - exact_filtered_energy: -0.38955310329705545
  - qpe_energy_estimate: None
```

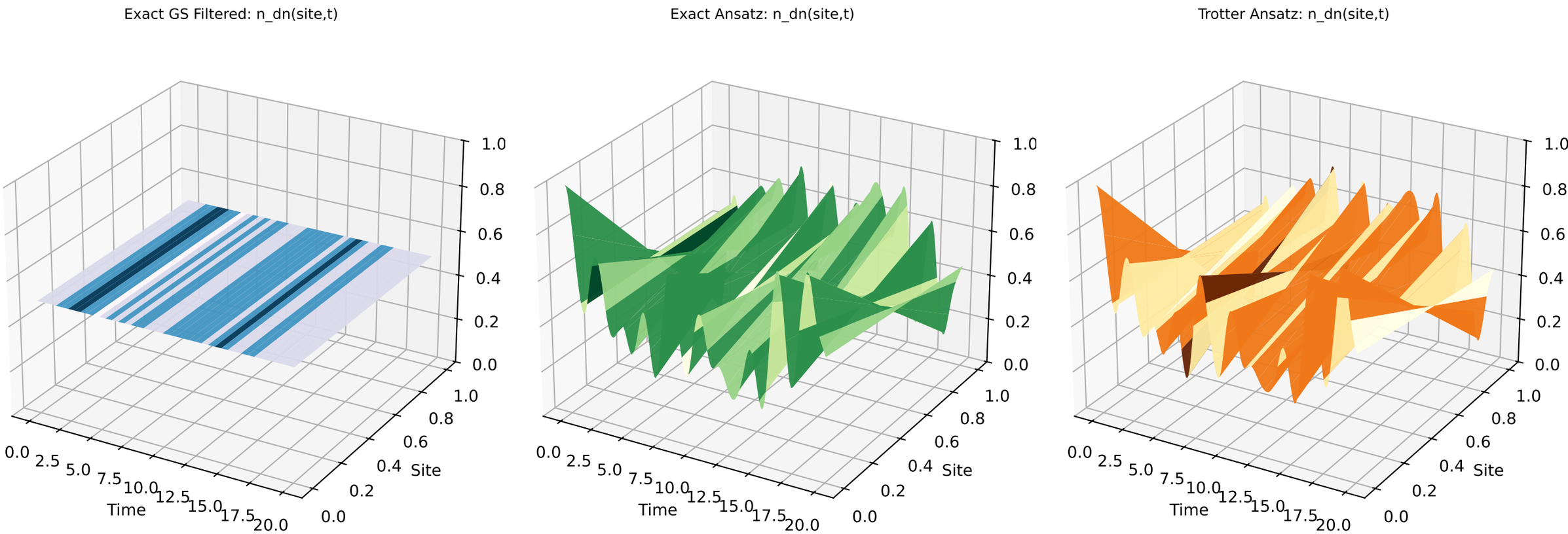
L=2 3D Densities (Total n)



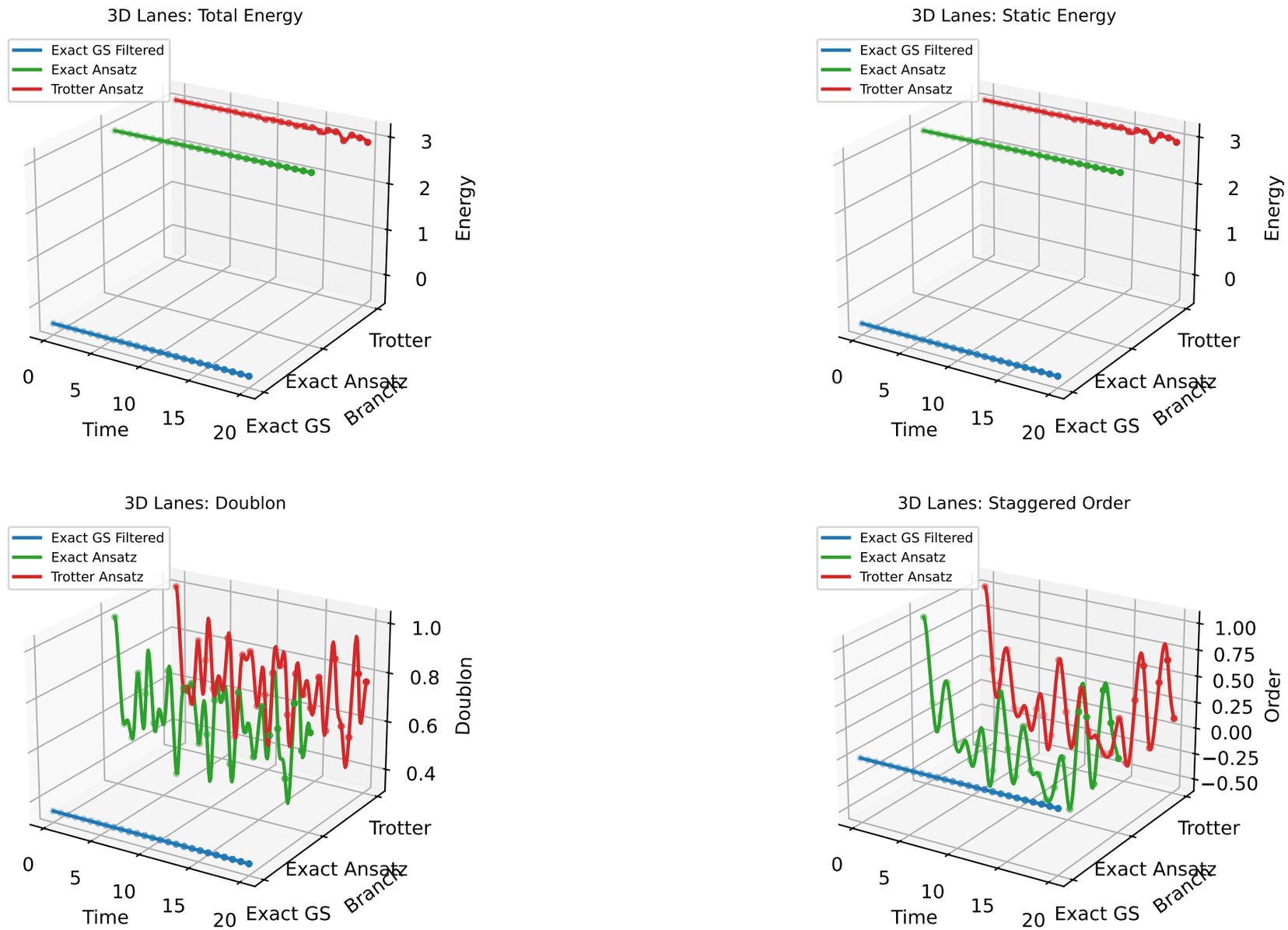
L=2 3D Densities (Spin-Up)

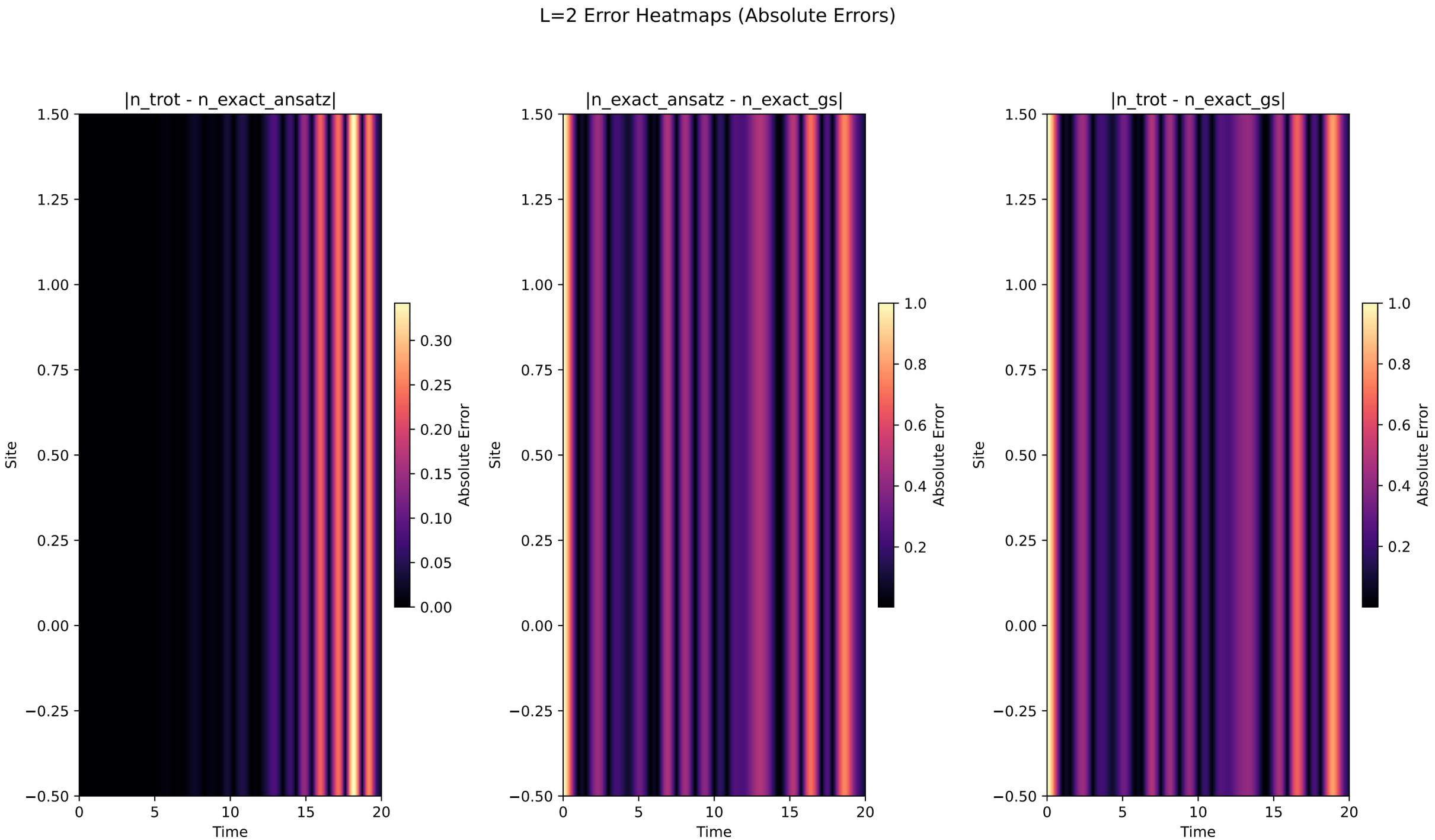


L=2 3D Densities (Spin-Down)

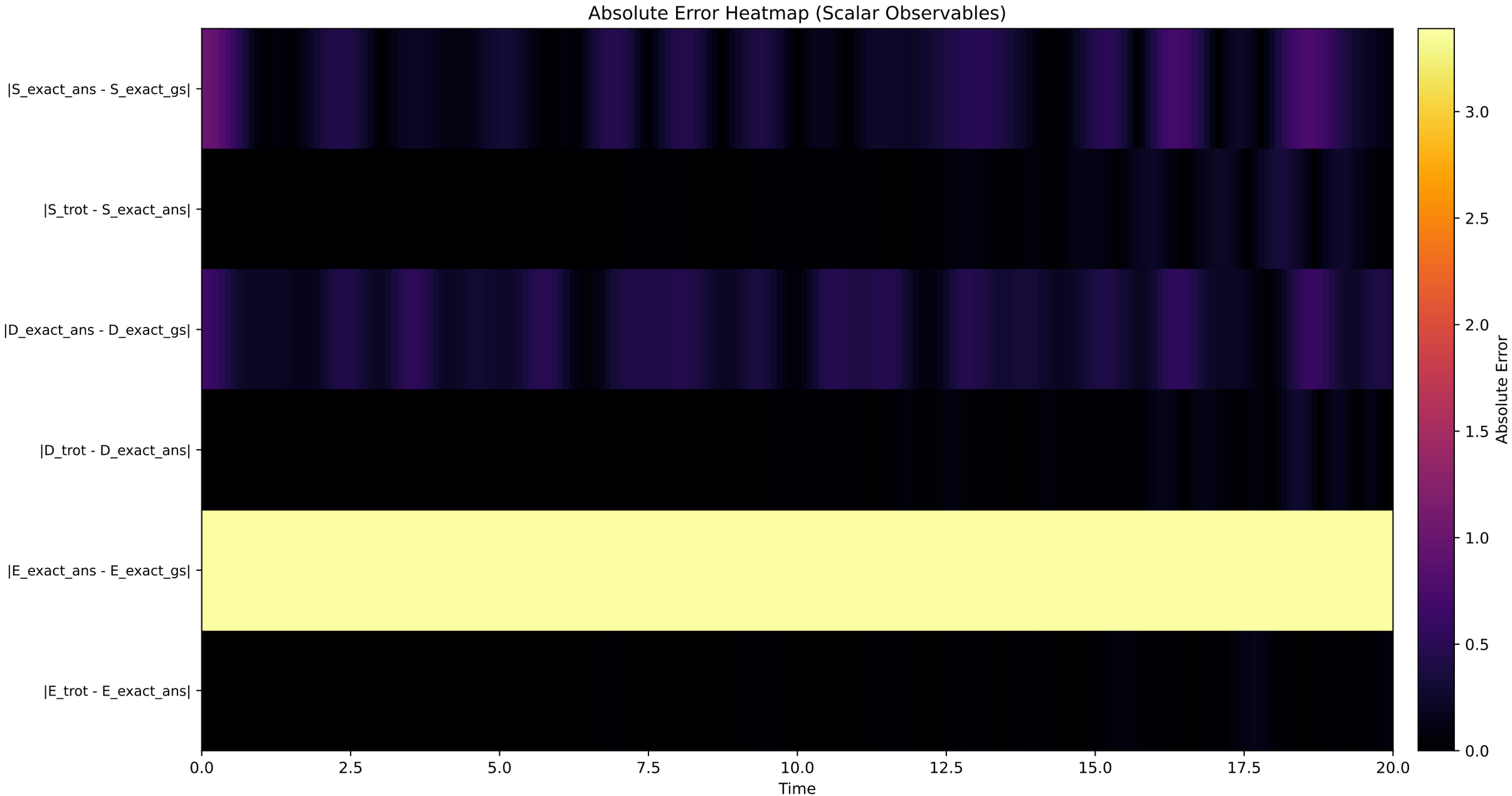


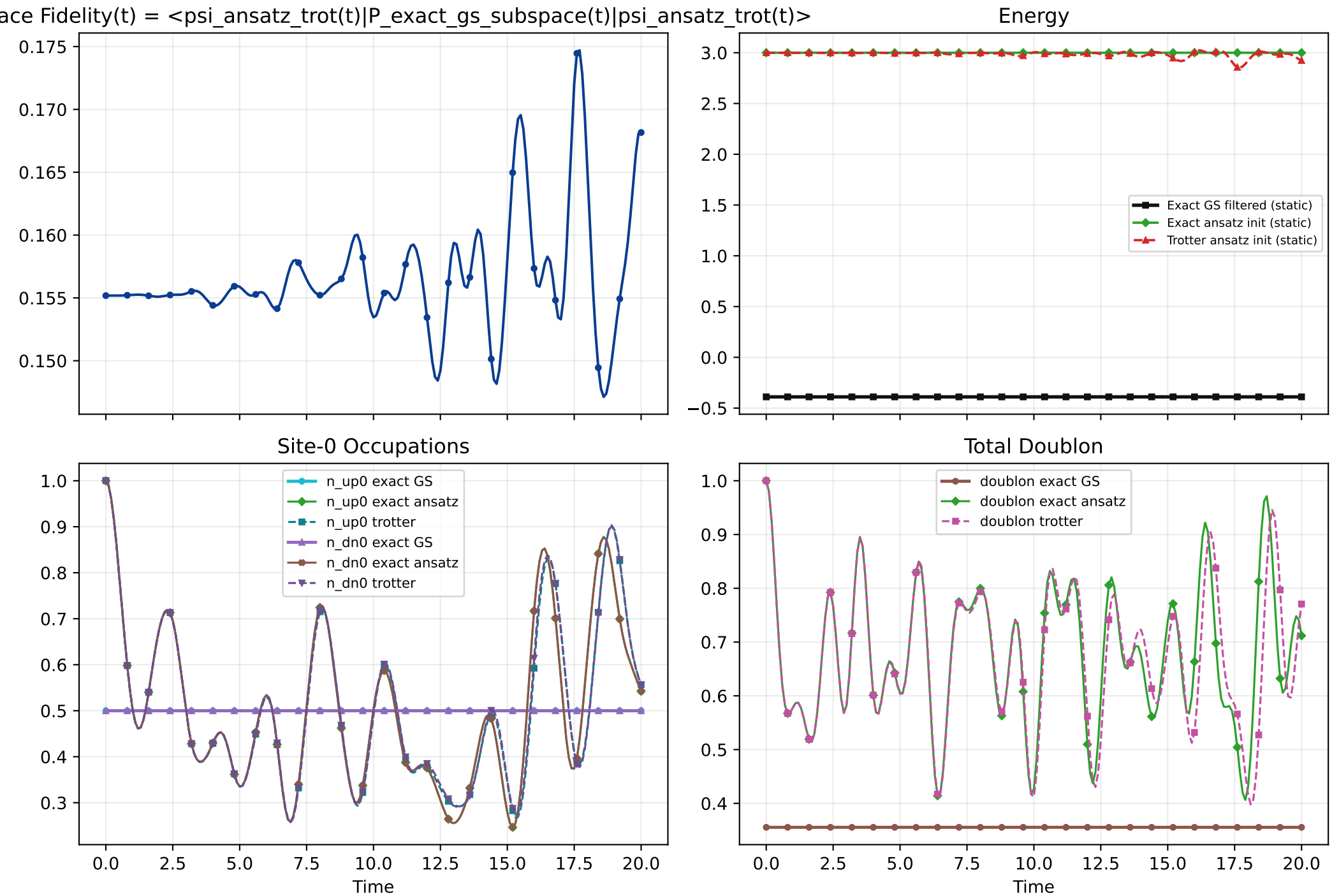
L=2 3D Scalar Observables (Three Evolutions)

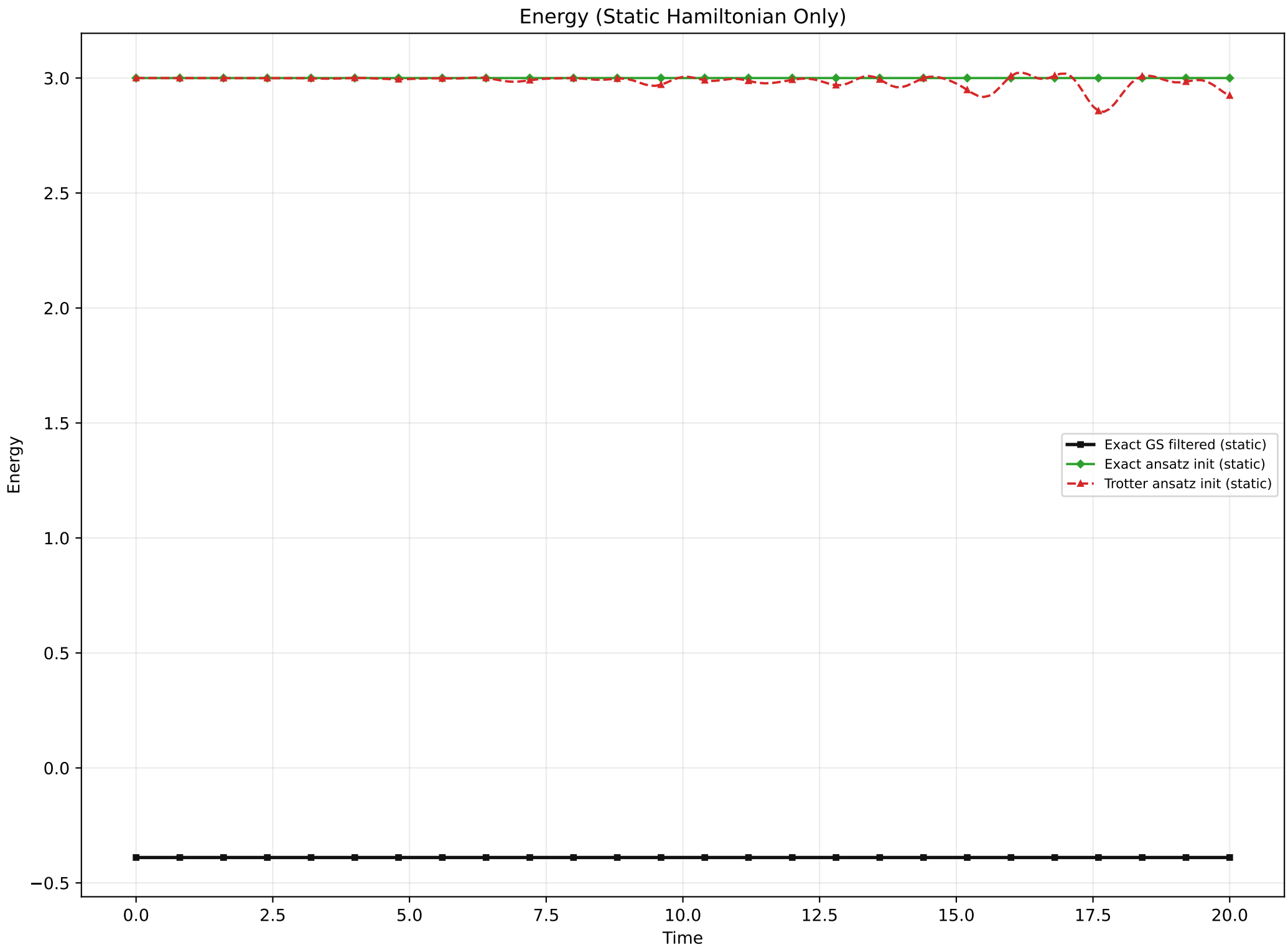


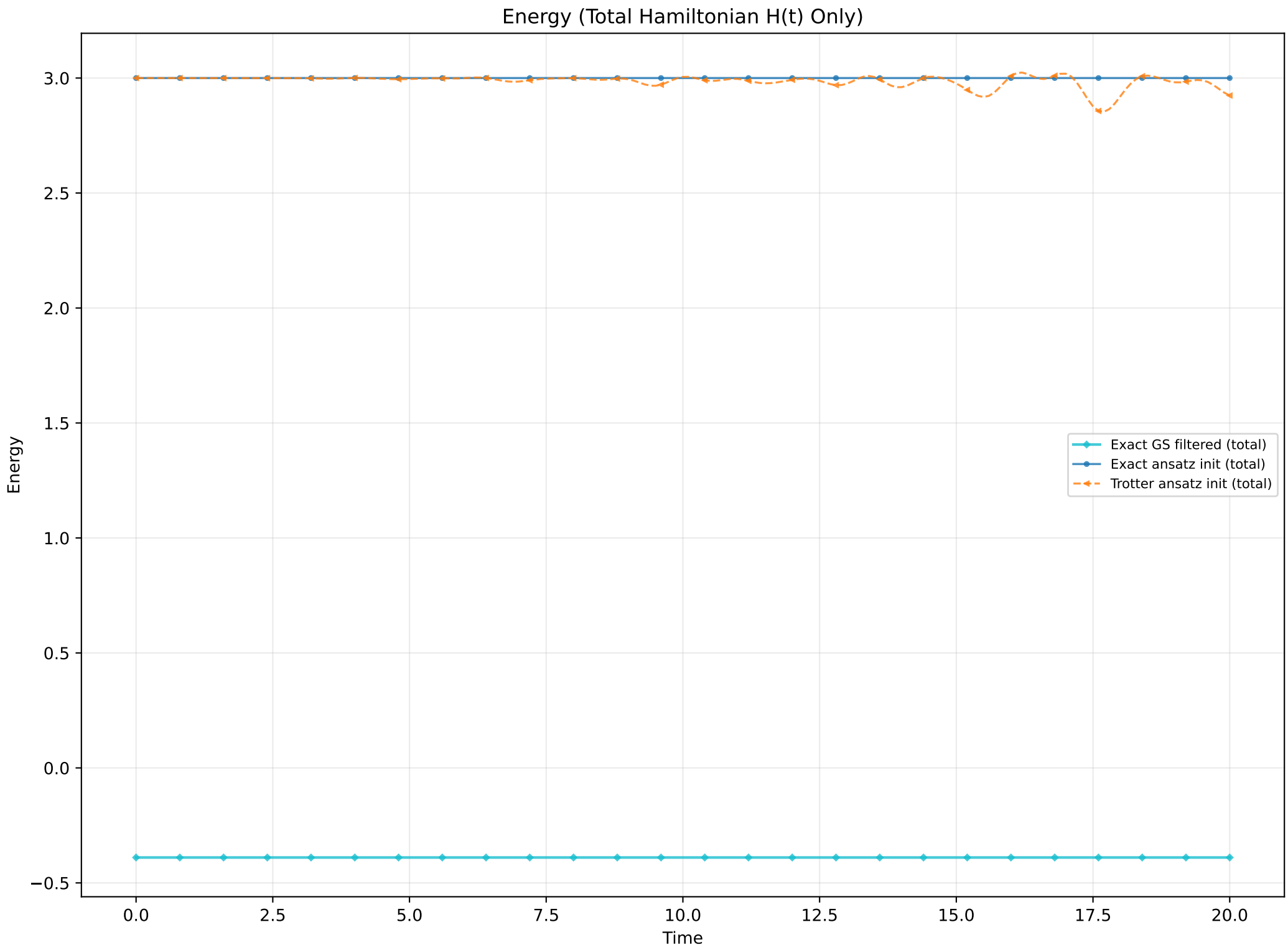

L=2 Error Heatmaps (Absolute Errors)
|n_trot - n_exact_ansatz|
|n_exact_ansatz - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
0
5
10
15
20
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1.0
0.8
0.6
0.4
0.2

L=2 Scalar Error Heatmap




Hardcoded Hubbard Pipeline: L=2
ace Fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>
Energy
Exact GS filtered (static)
Exact ansatz init (static)
Trotter ansatz init (static)
Site-0 Occupations
n_up0 exact GS
n_up0 exact ansatz
n_up0 trotter
n_dn0 exact GS
n_dn0 exact ansatz
n_dn0 trotter
Time
Total Doublon
doublon exact GS
doublon exact ansatz
doublon trotter
Time

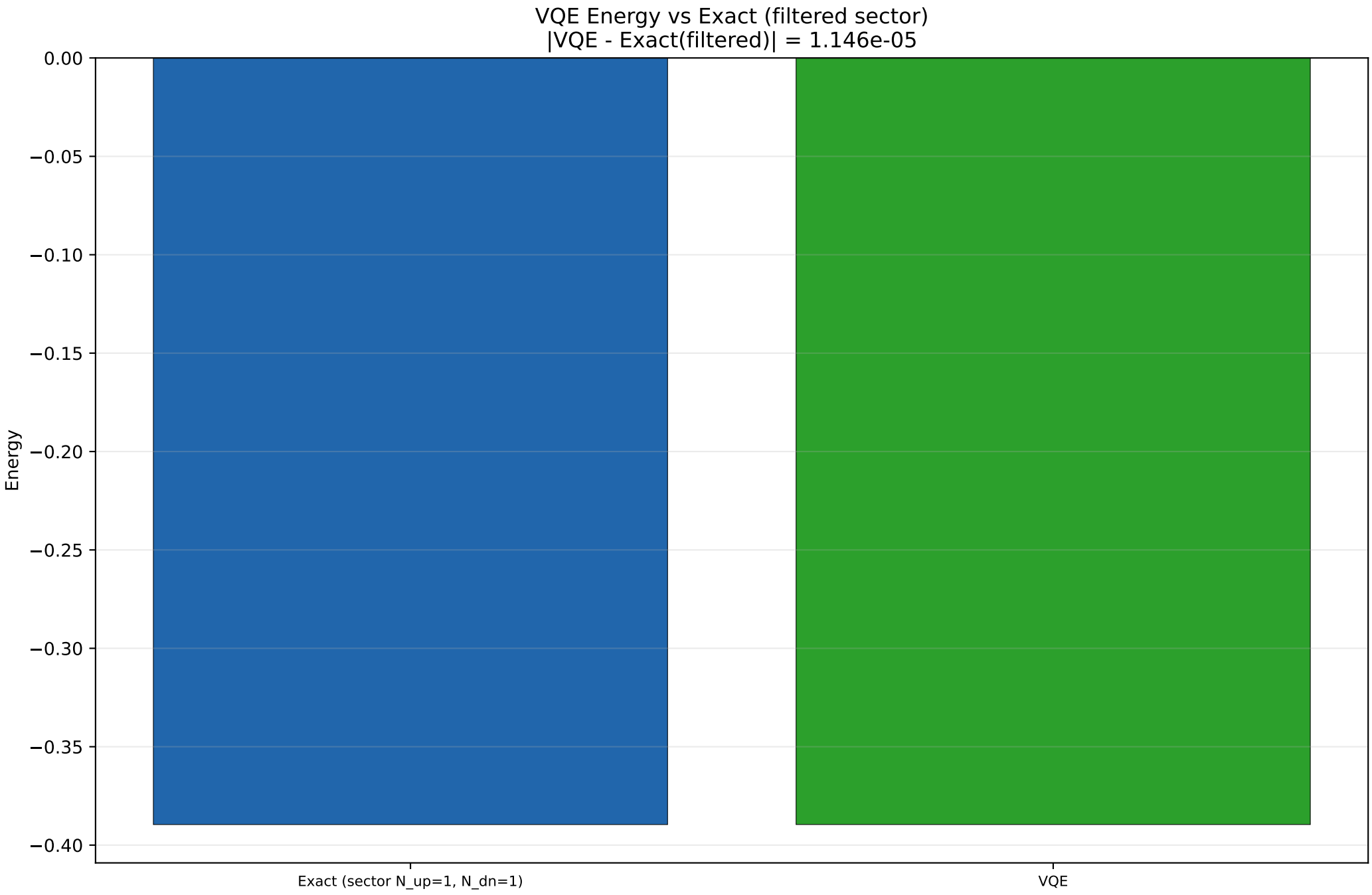

Energy (Static Hamiltonian Only)
Energy
Time
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Exact GS filtered (static)
Exact ansatz init (static)
Trotter ansatz init (static)


Energy (Total Hamiltonian H(t) Only)
Energy
Time
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Exact GS filtered (total)
Exact ansatz init (total)
Trotter ansatz init (total)

Appendix — L=2

This section contains supporting/redundant views and metadata.
Core non-redundant analysis appears earlier in the PDF.

VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

Ansatz:
  - hardcoded ansatz: hh_hva_tw
  - vqe method: hardcoded_hh_hva_termwise_statevector

Energy + Fidelity:
  - subspace_fidelity_at_t0: 0.1551829166136877
  - energy_t0_exact_gs: -0.3895531032970549
  - energy_t0_exact_ansatz: 2.9999999999999996
  - energy_t0_trotter: 3.0
  - ground_state_exact_energy_full_hilbert: -0.495507656605
  - ground_state_exact_energy_filtered: -0.38955310329705545
  - filtered_sector: {'n_up': 1, 'n_dn': 1}
  - vqe_energy: -0.3895416454240122
  - qpe_energy_estimate: None

Config:
  - initial_state_source: hf
  - exact_trajectory_label: Exact_Hardcode
  - exact_trajectory_method: python_matrix_eigendecomposition
  - fidelity_definition: fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>, where
    P_exact_gs_subspace(t) projects onto the time-evolved filtered-sector ground manifold selected by E <= E0 +
    tol.
  - hamiltonian_terms: 17
  - reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```